A RELATION WITH NON-NILPOTENT FORCING TERM OBSTRUCTS FINITE TOR DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. The identity theta^n(gamma_n(xi)) = g^n is introduced by 'Observe that' without displayed derivation; it is a two-to-three-line induction from the DP-compatibility axioms displayed in the statement of lemma-get-derivation. Declared dependency: lemma-get-derivation is not carded; recorded via external_references.

Allowed standard prerequisites (editorial).

- Tate resolution. lemma-tate-resolution (same chapter, named). Supplies R -> A -> S with the DP structure.
- Derivation theta. lemma-get-derivation (same chapter, named; NOT carded – build-gate exclusion). Supplies theta and its DP-compatibility axioms; declared as an external reference, not re-proved.
- Theta computation. lemma-compute-theta (same chapter, named). Supplies the class xi in H_2(A/IA) with theta(xi) = g in S/IS.

Source reference. Original lemma 09PT: sources/stacks-dpa/pinned-original-source.tex, source lines 1684–1684. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PS: sources/stacks-dpa/pinned-original-source.tex, source lines 1651–1651. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PN: sources/stacks-dpa/pinned-original-source.tex, source lines 1424–1424. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R' \to R$ be a surjection of Noetherian rings whose kernel has square zero and is generated by one element f. Let $S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. Let $\sum r_j f_j = 0$ be a relation in $R[x_1, \ldots, x_n]$. Assume that*

(1) *each r_j has coefficients in the annihilator I of f in R,*
(2) *for some lifts $r'_j, f'_j \in R'[x_1, \ldots, x_n]$ we have $\sum r'_j f'_j = gf$ where g is not nilpotent in S/IS.*

Then S does not have finite tor dimension over R (i.e., S is not a perfect R-algebra).

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Proof. Choose a Tate resolution $R \to A \to S$ as in Lemma 09PN. Let $\xi \in H_2(A/IA)$ and $\theta : A/IA \to A/IA$ be the element and derivation found in Lemmas 09PS and 09PT. Observe that

$$\theta^n(\gamma_n(\xi)) = g^n$$

in $H_0(A/IA) = S/IS$. Hence if g is not nilpotent in S/IS, then ξ^n is nonzero in $H_{2n}(A/IA)$ for all $n > 0$. Since $H_{2n}(A/IA) = \mathrm{Tor}_{2n}^R(S, R/I)$ we conclude. $\square$